

On Sampling of Band-Limited Signals Associated With the Linear Canonical Transform

Ran Tao, *Senior Member, IEEE*, Bing-Zhao Li, Yue Wang, and George Kwamina Aggrey

Abstract—Sampling is one of the fundamental topics in the signal processing community. Theorems proposed under this topic form the bridge between the continuous-time signals and discrete-time signals. Several sampling theorems, which aid in the reconstruction of signals in the linear canonical transform (LCT) domain, have been proposed in the literature. However, two main practical issues associated with the sampling of the LCT still remain unresolved. The first one relates to the reconstruction of the original signal from nonuniform samples and the other issue relates to the fact that only a finite number of samples are available practically. Focusing on these issues, this paper seeks to address the above from the LCT point of view. First, we extend several previously developed theorems for signals band-limited in the Fourier domain to signals band-limited in the LCT domain, followed by the derivation of the reconstruction formulas for finite uniform or recurrent nonuniform sampling points associated with the LCT. Simulation results and the potential applications of the theorem are also proposed.

Index Terms—Chirp-periodicity, linear canonical transform, nonuniform sampling, signal reconstruction, uniform sampling.

I. INTRODUCTION

NONSTATIONARY signal processing plays a fundamental role in modern digital signal processing and evokes considerable interest in the literature. Many useful tools such as the short time Fourier transform, the wavelet transform [1], the fractional Fourier transform (FrFT) and the linear canonical transform (LCT) [2]–[4] have been used in processing nonstationary signals. The LCT [3]–[10], which was introduced during the 1970s with four parameters, has been proven to be one of the most powerful tools for nonstationary signal processing. Although it is not very well known, its special cases are widely used in various fields under different names [12]. The well-known signal processing operations, such as the Fourier transform (FT), the FrFT [3], [4], the Fresnel transform [8], and the scaling operations are all special cases of the LCT. The LCT has also found many applications in the solution of

optical systems [9]–[12], radar system analysis, gradient-index (GRIN) [13] medium system analysis, filter design, pattern recognition, holographic three-dimensional television and many others [7]–[12]. Therefore, understanding the LCT may help to gain more insight into its special cases and to carry the knowledge gained from one subject to others [12].

One of the most fundamental concepts of digital signal processing is the sampling process, which serves as a bridge between the continuous physical signals and discrete signals. The expansion of the classical uniform sampling theorem for the band-limited or time-limited signal in the LCT domain has recently been studied [14]–[16]. These sampling theorems establish the fact that a band-limited or time-limited continuous signal in the LCT domain can be completely reconstructed by a set of equidistantly spaced signal samples. However, there are two pending issues associated with the sampling of the band-limited signal in the LCT domain.

The first one relates to the reconstruction of the original signal from nonuniform samples in the LCT domain. To the best of our knowledge, there are no results published associated with the nonuniform sampling in the LCT domain. However, there are several apparent reasons justifying the need for nonuniform sampling [17]–[26]. First, nonuniform or irregular sampling measurements do occur naturally in several applications, including optics [18], tomography [19], [20], frequency modulation (FM), phase, and delta or pulse position demodulation [21], [22]. Second, it is often difficult to sample uniformly certain types of data, such as astronomical or geophysical data. Third, there are several factors that may accidentally lead to irregularly distributed sets of samples: jitter, incorrect samples due to noise or clipping, lost or delayed packets in packet-oriented telecommunication systems, data losses due to channel erasures, and others [23], [24]. Fourth, some physical problems have simple description using a nonuniform grid [25].

The other issue associated with the sampling formulas in the LCT domain is the numerical implementation of the reconstruction methods using a digital computer. In other words, in most practical applications we are given only finite number of samples [32], [33], which may not be suitable for the perfect reconstruction of a signal.

For the above mentioned nonuniform sampling cases in the traditional Fourier domain, it is well established [26]–[31] that a band-limited continuous-time signal $x(t)$ can be reconstructed from its samples at a set of sampling times $\{t_n\}$ if the average sampling period is smaller than the Nyquist period associated with the Fourier transform, where the average sampling period is defined as $\lim_{n \rightarrow \infty} t_n/n$. When applying these common sampling theorems [26]–[31] to signals non-band-limited in Fourier domain, however band-limited in the LCT domain, may lead

Manuscript received January 9, 2008; revised June 26, 2008. First published August 8, 2008; current version published October 15, 2008. The associate editor coordinating the review of this paper and approving it for publication was Prof. Yonina C. Eldar. This work was supported in part by the National Natural Science Foundation of China (10671013), the National Science Fund for Distinguished Young Scholars (60625104) and also sponsored in part by the Basic Science Foundation of Beijing Institute of Technology.

R. Tao, Y. Wang and G. K. Aggrey are with the Electronic Engineering Department, Beijing Institute of Technology, Beijing, China, 100081 (e-mail: rantao@bit.edu.cn).

B.-Z. Li is with the Electronic Engineering Department and the Mathematics Department, Beijing Institute of Technology, Beijing, China, 100081 (e-mail: li_bingzhao@bit.edu.cn).

Digital Object Identifier 10.1109/TSP.2008.929333

to wrong (or at least suboptimal) conclusions. In order to overcome these problems, we investigate this issue for some special nonuniform sampling models in the LCT domain and derive the nonuniform sampling formulas for signals band-limited in the LCT domain in this draft.

The paper is organized as follows. In Section II, the LCT along with some properties is introduced. The nonuniform sampling formulas associated with the LCT are derived in Section III. The reconstruction formulas for chirp-periodic band-limited signal in the LCT domain using only finite samples are proposed in Section IV. Simulation results and potential applications are presented in Section V. This paper is concluded in Section VI.

II. PRELIMINARY

A. The Linear Canonical Transform

The LCT with parameters $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a signal $f(t)$ is defined as [3], [5] (1) shown at the bottom of the page, where

$$\mathbf{A} \equiv \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ and } \det(\mathbf{A}) = ad - bc = 1.$$

In this paper, we restrict ourselves to the class of the LCT with real parameters $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, in such cases the LCT operator is unitary in $L^2(R)$, and the inverse LCT is given by the LCT having parameters

$$\mathbf{A}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

In other words, we can reconstruct the original signal $f(t)$ from $L_f^{\mathbf{A}}(u)$ via (2), shown at the bottom of the page.

It should be noted that, when $b = 0$, the LCT of a signal is essentially a chirp multiplication and it is of no particular interest to our objective in this work, so it will not be discussed in this paper, and without loss of generality, we assume $b > 0$ in this paper. It can be verified [12] that the LCT with parameters $\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ reduces to FrFT and to FT when $\theta = \pi/2$. The LCT also reduces to the Fresnel transform if $\begin{pmatrix} a & b \\ c & d \end{pmatrix} =$

$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$. Multiplication by a Gaussian or chirp function is obtained with an $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}$. The scaling operator can be viewed as a special case of the LCT with $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d^{-1} & 0 \\ 0 & d \end{pmatrix}$. For further details about the definition and properties of the LCT, readers can refer to [3]–[12].

A signal $f(t)$ is said to be band-limited to $\Omega_{\mathbf{A}}$ in the LCT sense with parameter $\mathbf{A}$ if

$$L_f^{\mathbf{A}}(u) = 0 \text{ for } |u| > \Omega_{\mathbf{A}} \quad (3)$$

where $\Omega_{\mathbf{A}}$ is the bandwidth of the signal $f(t)$ in the LCT domain. A band-limited signal in the LCT domain has the following properties.

Lemma 1: Assume a signal $f(t)$ is band-limited to $\Omega_{\mathbf{A}}$ in the LCT domain with parameters $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $b > 0$. Let

$$g(t) = \int_{-\Omega_{\mathbf{A}}}^{\Omega_{\mathbf{A}}} L_f^{\mathbf{A}}(u) e^{-jd/2bu^2 + j1/but} du \quad (4)$$

then $g(t)$ is a signal band-limited to $1/b\Omega_{\mathbf{A}}$ in the Fourier domain.

Proof: The result follows from the definition of band-limited signal in the FT domain [15]. ■

Lemma 2: If a nonzero signal $f(t)$ is band-limited in the LCT domain with parameter $\mathbf{A}_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$, then $f(t)$ cannot be band-limited with another parameter $\mathbf{A}_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$, satisfying the relationship $a_1/b_1 \neq a_2/b_2$.

Proof: The results can be derived easily by following Xia's method [34] associated with the FrFT. ■

It can be deduced from Lemma 2 that if a signal is non-band-limited in the Fourier domain, then it can be band-limited in the LCT domain. In [35], similar ideas have been presented; the authors derived the cases for non-band-limited signals in the Fourier domain. However they are band-limited in the LCT domain. Therefore, the common (Fourier) sampling theorem may mislead and the traditional non-band-limited signals processing problems in the Fourier domain can be solved in the LCT domain.

$$L_f^{\mathbf{A}}(u) = L_f^{\mathbf{A}}[f(t)](u) = \begin{cases} \sqrt{\frac{1}{j2\pi b}} \int_{-\infty}^{+\infty} f(t) e^{j1/2[a/bt^2 - (2/b)tu + d/bu^2]} dt, & b \neq 0 \\ \sqrt{d} e^{j1/2cd u^2} f(du), & b = 0 \end{cases} \quad (1)$$

$$f(t) = \begin{cases} \sqrt{\frac{1}{-j2\pi b}} \int_{-\infty}^{+\infty} L_f^{\mathbf{A}}(u) e^{j1/2[-d/bu^2 + (2/b)tu - a/bt^2]} du, & b \neq 0 \\ \sqrt{a} e^{-j1/2cat^2} f(at), & b = 0 \end{cases} \quad (2)$$

B. Uniform Sampling Theorem for Band-Limited Signal in the LCT Domain

The following uniform sampling formula associated with the LCT has been derived by Stern [14], [16] and Li [15].

Lemma 3: Assume $f(t)$ to be band-limited to Ω_A in the LCT domain with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $b > 0$; then the sampling theorem expansion for the signal $f(t)$ can be expressed as

$$f(t) = e^{-ja/2bt^2} \sum_{n=-\infty}^{+\infty} f(nT) e^{ja/2b(nT)^2} \frac{\sin \left[\frac{\Omega_A(t-nT)}{b} \right]}{\frac{\Omega_A(t-nT)}{b}} \quad (5)$$

where T is the sampling period and satisfies $T = \pi b / \Omega_A$; and the Nyquist rate of sampling theorem associated with the LCT is $f_s = \Omega_A / \pi b$.

Proof: See [14]–[16]. ■

Equation (5) establishes the relationship between the signal samples and the original signal, that is to say, we can reconstruct the original signal or the LCT of a band-limited signal in the LCT domain from the uniform sampling points of the signal, provided the sampling interval satisfies the uniform sampling conditions of Lemma 3. However, when applying this formula in practical situations, two problems as mentioned in Section I may be encountered. Recall that, the first problem is concerned with the nonuniform sampling signal processing, and the other problem is the numerical implementation of the existing reconstruction methods from finite sampling points. Focusing on these problems, this manuscript seeks to address the above mentioned from the LCT point of view in Sections III–V.

III. THE NONUNIFORM SAMPLING FORMULAS

A. The Recurrent Nonuniform Sampling

The recurrent nonuniform sampling model is a simple and interesting nonuniform sampling distribution. It arises in a broad range of applications. For example, we might consider converting a continuous-time signal to a discrete-time signal using a series of analog-to-digital (A/D) converters, each operating at a rate lower than the Nyquist rate, such that the average sampling rate is equal to the Nyquist rate. This may be beneficial in applications where high-rate A/D converters are required. Typically, the cost and complexity of a converter will increase (more than linearly) with the rate. In such cases, we can benefit from converting a continuous-time signal to a discrete-time signal using A/D converters, each operating at one N th of the Nyquist rate [29], [30], [37]. Since the converters are typically not synchronized, the resulting discrete-time signal is a combination of sequences of uniform samples, where each sequence corresponds to samples at one N th of the Nyquist rate of a time delayed version of the continuous-time signal. Thus, the resulting discrete-time signal corresponds to recurrent nonuniform samples of the continuous-time signal. Fig. 1. shows the model of a recurrent nonuniform sampling.

Existing results about signal reconstruction from recurrent nonuniform sampling points are mostly limited to the band-limited signals in the Fourier domain, [26]–[30], [36]–[41]; but applying these methods to non-band-limited signals may lead to wrong (or at least suboptimal) conclusions about signals and systems that have band-limited LCT. It can be concluded from

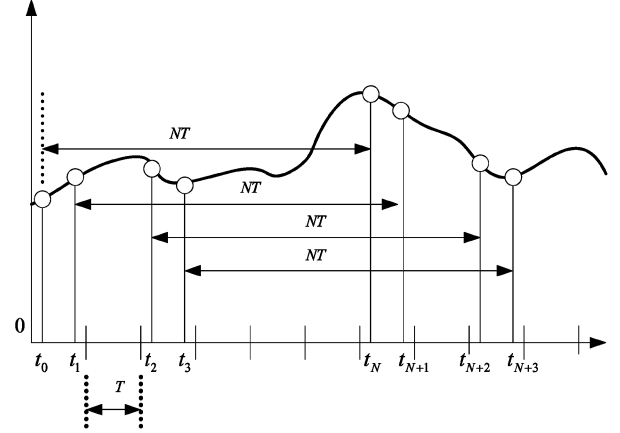


Fig. 1. Recurrent nonuniform sampling model [29].

Lemma 2, that a non-band-limited signal in the Fourier domain can be band-limited in a certain LCT domain. Thus, the classical band-limited signal processing method can be used to analyze signals in the LCT domain.

In order to derive the sampling formula for recurrent nonuniform sampling points associated with the LCT, the sampling points of a Ω_A band-limited signal in the LCT domain that satisfy the LCT sampling condition [14], [15] are divided into groups of N points with each group having a recurrent period of $Nb\pi/\Omega_A$ seconds. In one period the points are denoted by t_p , $p = 1, 2, \dots, N$. The complete set of sample points can be written as $\tau_{pm} = t_p + mNb\pi/\Omega_A = t_p + mNT$, $m = \dots, -1, 0, 1, \dots$. For this kind of recurrent nonuniform sampling distribution the following theorem is valid.

Theorem 1: If a signal $f(t)$ is Ω_A band-limited signal in the LCT domain such that $L_f^A(u) = 0$ for $|u| > \Omega_A$, then the original signal $f(t)$ can be reconstructed by the recurrent nonuniformly sampled points $\tau_{pm} = t_p + mNT$, $p = 1, 2, \dots, N$, $m = \dots, -1, 0, 1, \dots$, and the interpolation formula can be presented as follows:

$$f(t) = e^{-ja/2bt^2} \sum_{m=-\infty}^{\infty} \sum_{p=1}^N f(\tau_{pm}) e^{ja/2b\tau_{pm}^2} \psi_{pm}(t) \quad (6)$$

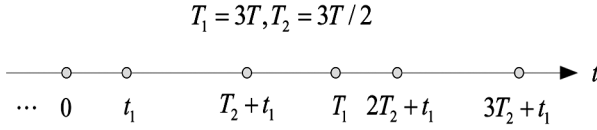
where

$$\psi_{pm}(t) = \frac{\prod_{q=1}^N \sin \frac{\Omega_A}{bN}(t - t_q)}{\prod_{\substack{q=1 \\ q \neq p}}^N \sin \frac{\Omega_A}{bN}(t_p - t_q)} \frac{(-1)^{mN}}{\frac{\Omega_A}{bN} \left(t - t_p - mN \frac{\pi b}{\Omega_A} \right)}. \quad (7)$$

Proof: Since $f(t)$ is a Ω_A band-limited signal in the LCT domain, let $g(t) = \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-jd/2bu^2 + j1/but} du$, then $g(t)$ is a $1/b\Omega_A$ band-limited signal in the Fourier domain from Lemma 1. The final result is obtained by applying the classical result derived in [29] to $g(t)$. ■

B. N th-Order Recurrent Nonuniform Sampling

In this subsection, we consider a subclass of recurrent nonuniform sampling, in which the sampling points in each period can be further divided into groups with a common inner period [27].

Fig. 2. N th-order recurrent nonuniform sampling model [27].

Specifically, the samples consist of N sequences of uniform samples taken at N rates, i.e., the set of sampling points are

$$t_p + nT_p, \quad p = 1, \dots, N, \quad n \in (-\infty, \infty) \quad (8)$$

where

$$\sum_{p=1}^N \frac{1}{T_p} = \frac{1}{T} = \frac{\Omega_A}{b\pi} \quad (9)$$

so that the average sampling rate is the Nyquist rate in the LCT domain. An example of a sampling distribution for the case $N = 2$, with $T_1 = 3T$, $T_2 = 3T/2$ is depicted in Fig. 2.

For a deeper understanding of this kind of nonuniform sampling model and the filterbank reconstruction methods in the Fourier domain, the reader is referred to [27]. This kind of nonuniform sampling may arise in a similar context as the recurrent nonuniform sampling, namely, when converting a continuous-time signal to a discrete-time signal using a series of A/D converters, each operating at a different rate, such that the average sampling rate is equal to the Nyquist rate. The extension of the sampling formula from the N th-order recurrent nonuniform sampling points associated with the LCT can be derived as following.

Theorem 2: If $f(t)$ is a band-limited signal in the LCT domain such that $L_f^A(u) = 0$ for $|u| > \Omega_A$, then the original signal $f(t)$ can be reconstructed by the N th-order recurrent nonuniformly sampled points $\tau_p = t_p + nT_p$, and the interpolation formula can be presented as follows:

$$f(t) = e^{-ja/2bt^2} \sum_{p=1}^N \sum_{n=-\infty}^{\infty} f(\tau_p) e^{ja/2b\tau_p^2} \times \frac{T_p \prod_{q=1}^N \sin\left(\frac{\pi(t-t_q)}{T_q}\right)}{\pi(-1)^n \prod_{q=1, q \neq p}^N \sin\left(\frac{\pi(t_p-t_q)}{T_q} + n\pi \frac{T_p}{T_q}\right) (t - nT_p - t_p)} \quad (10)$$

Proof: Similar to the proof of Theorem 1. ■

C. Nonuniform Sampling Due to Migration of a Finite Number of Uniform Samples

Besides the recurrent nonuniform sampling model, one of the simplest nonuniform sampling distribution is obtained from the migration of finite number of sample points in a uniform distribution. Yen considered this type of nonuniform sampling pattern and deduced a classical reconstruction theorem for band-limited signals in the traditional Fourier domain [39], however, when applying his theorem to signals non-band-limited in the

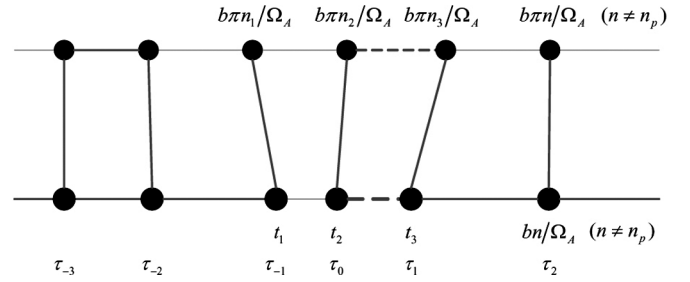


Fig. 3. Nonuniform sampling model [39].

Fourier domain but, band-limited in the LCT domain, may lead to wrong (or at least suboptimal) conclusions. In order to solve this problem, the generalized nonuniform sampling pattern associated with the LCT is derived in this subsection.

Similar to the nonuniform sampling model [39], this kind of nonuniform sampling model can be modeled as shown in Fig. 3. It shows N uniform sample points of Ω_A band-limited signal associated with the LCT located at $t = b\pi n_p/\Omega_A$, where n_p with $p = 1, 2, \dots, N$ are N distinct integers, have been migrated to N new positions $t = t_p$, and $\Omega_A t_p/\pi b$ is not an integer.

For this kind of nonuniform sampling model, the following nonuniform sampling theorem associated with the LCT is valid.

Theorem 3: If a signal $f(t)$ is Ω_A band-limited in the LCT domain such that $L_f^A(u) = 0$ for $|u| > \Omega_A$, then the original signal $f(t)$ can be reconstructed from the nonuniformly sampled points τ_m and the interpolation formula can be presented as follows:

$$f(t) = e^{-ja/2bt^2} \sum_{m=-\infty}^{\infty} f(\tau_m) e^{ja/2b\tau_m^2} \psi_m(t) \quad (11)$$

where [see (12) at the bottom of the next page].

Proof: Since $f(t)$ is a Ω_A band-limited signal in the LCT domain, let $g(t) = \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-jd/2bu^2 + j1/but} du$, then from Lemma 1, $g(t)$ is a $1/b\Omega_A$ band-limited signal in the Fourier domain. The classical result derived in [39] is applied to $g(t)$ resulting in

$$g(t) = \sum_{m=-\infty}^{\infty} g(\tau_m) \psi_m(t) \quad (13)$$

Since

$$f(t) = \sqrt{\frac{1}{-j2\pi b}} e^{-ja/2bt^2} g(t). \quad (14)$$

The final result can therefore be proven from (10)–(11). ■

D. The General Nonuniform Sampling Model

Besides the above mentioned special nonuniform sampling model, we consider a more general nonuniform sampling model in this subsection. Suppose that a Ω_A band-limited signal $f(t)$ in the LCT domain is sampled nonuniformly in the time domain. Then the original signal $f(t)$ can be reconstructed from these nonuniform sampling points from the following theorem if a stronger constrain is introduced.

Theorem 4: Let a signal $f(t)$ be Ω_A band-limited signal in the LCT domain with parameters $\mathbf{A}$, such that $L_f^{\mathbf{A}}(u) = 0$ for $|u| > \Omega_A$, then the original signal $f(t)$ is uniquely determined by its samples $f(t_n)$ if

$$\left| t_n - n \frac{b\pi}{\Omega_A} \right| \leq D < \frac{b\pi}{4\Omega_A} \quad (15)$$

and the reconstruction formula is given by

$$f(t) = e^{-ja/2bt^2} \sum_{n=-\infty}^{\infty} f(t_n) e^{-ja/2bt_n^2} \frac{G(t)}{G'(t_n)(t-t_n)} \quad (16)$$

where

$$G(t) = e^{\alpha t(t-t_0)} \prod_{n \neq 0} \left(\frac{1-t}{t_n} \right) e^{t/t_n};$$

$$\alpha = \sum_{n \neq 0} \frac{1}{t_n} \quad D \in \mathbb{R};$$

and $G'(t_n)$ is the derivative of $G(t)$ evaluated at $t = t_n$, and $t_n = 0$ for some n , when $t_0 = 0$.

Proof: Similar to the proof of Theorem 2 and using the classical results of [38]. ■

It can be shown that the results of the above derived theorems reduce to the classical nonuniform sampling results in the Fourier domain when the parameters of the LCT reduce to $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The nonuniform sampling theorems in the FrFT domain can also be derived just by letting $\mathbf{A} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$. Furthermore, the results derived above can also be used for nonstationary signal reconstruction from nonuniformly sampled points, such as chirp signal processing.

IV. THE SIGNAL RECONSTRUCTION FROM FINITE SAMPLES

The reconstruction formula for band-limited signals from nonuniform samples associated with the LCT has been deduced in Section III, but the numerical implementation of the above reconstruction methods on a digital computer has not been possible, i.e., there is an infinite number of sampling points needed to perfectly reconstruct a signal. Moreover, in most practical applications we are given only a finite number of samples, which makes it impossible to reconstruct a signal perfectly. To overcome these limitations, we first investigate the truncation errors of band-limited signals in the LCT domain

from finite uniform samples, and then derive the reconstruction formulas for a small class of signals with discrete band-limited LCT from finite uniform or recurrent nonuniform samples.

A. Error Analysis of Band-Limited Signal in the LCT Domain

There are several types of errors which might influence the accuracy of the reconstruction of a signal from its finite samples, for example, the truncation error $T_N f$, which occur naturally in applications, arise if only a finite number N of samples are taken into account instead of the infinitely many of the sampling series for band-limited signal $f(t)$. The definition of the truncation error associated with the LCT is similar to that of the definition in the Fourier domain and can be presented as follows:

$$T_N f = f(t) - e^{-ja/2bt^2} \sum_{n=-N}^{+N} f(nT) e^{ja/2b(nT)^2} \frac{\sin \left[\frac{\Omega_A(t-nT)}{b} \right]}{\frac{\Omega_A(t-nT)}{b}}$$

$$= e^{-ja/2bt^2} \sum_{|n| > N} f(nT) e^{ja/2b(nT)^2} \frac{\sin \left[\frac{\Omega_A(t-nT)}{b} \right]}{\frac{\Omega_A(t-nT)}{b}}. \quad (17)$$

Besides the signal $f(t)$ being band-limited in the LCT domain, if some extra conditions are introduced, then the truncation error associated with the LCT can be obtained as follows.

Theorem 5: If $f(t)$ is a band-limited signal in the LCT domain and its LCT has a support smaller than $r\Omega_A$, $0 < r < 1$, let $M = \max_{-\infty < t < +\infty} |f(t)|$, then the truncation error $T_N f$ given by (17) satisfies

$$|T_N f| \leq \frac{4M}{\pi^2 N(1-r)}. \quad (18)$$

Proof: Since $f(t)$ is a band-limited signal in the LCT domain with a support smaller than $r\Omega_A$, let $g(t) = \int_{-\Omega_A}^{\Omega_A} L_f^{\mathbf{A}}(u) e^{-jd/2bu^2 + j1/but} du$, then $g(t)$ is $r/b\Omega_A$ band-limited signal in the Fourier domain. Equation (17) can then be rewritten as follows:

$$T_N f = f(t) - e^{-ja/2bt^2} \sum_{n=-N}^{+N} f(nT) e^{ja/2b(nT)^2} \times \frac{\sin \left[\frac{\Omega_A(t-nT)}{b} \right]}{\frac{\Omega_A(t-nT)}{b}}$$

$$= \sqrt{\frac{1}{-j2\pi b}} e^{-ja/2bt^2} T_N g. \quad (19)$$

$$\psi_m(t) = \begin{cases} \frac{\prod_{q=1}^N (t-t_q) \prod_{q=1}^N (\frac{n\pi b}{\Omega_A} - \frac{n_q \pi b}{\Omega_A})}{\prod_{q=1}^N (t - \frac{n_q \pi b}{\Omega_A}) \prod_{q=1}^N (\frac{n\pi b}{\Omega_A} - t_q)} \cdot \frac{(-1)^n b \sin[(\frac{1}{b}\Omega_A)t]}{(\Omega_A t - n\pi b)}, & \tau_m = \frac{n\pi b}{\Omega_A} \neq \frac{n_q \pi b}{\Omega_A} \\ \frac{\prod_{q=1, q \neq p}^N (t-t_q) \prod_{q=1}^N (t_p - \frac{n_q \pi b}{\Omega_A})}{\prod_{q=1}^N (t - \frac{n_q \pi b}{\Omega_A}) \prod_{q=1, q \neq p}^N (t_p - t_q)} \cdot \frac{\sin(\frac{1}{b}\Omega_A t)}{\sin(\frac{1}{b}\Omega_A t_p)}, & \tau_m = t_p. \end{cases} \quad (12)$$

Because

$$\begin{aligned} & \max_{-\infty < t < +\infty} |g(t)| \\ &= \max_{-\infty < t < +\infty} \left| f(t) \sqrt{-j2\pi b} e^{j(a/2b)t^2} \right| \\ &= M\sqrt{2\pi b} \end{aligned} \quad (20)$$

and the truncation error of the signal $g(t)$ can be deduced from the classical results [40], [41] as

$$|T_N g| \leq \frac{4\sqrt{2\pi b}M}{\pi^2 N(1-r)}. \quad (21)$$

The final results can be obtained through (19)–(21). ■

B. Chirp-Periodic Uniform Sampling Signals

In part A of Section IV, we deduced the truncation error for a general band-limited signal in the LCT domain from a finite number of signal samples by imposing some extra conditions on the signal $f(t)$. In this subsection we consider a small class of signals, namely signals that are chirp periodic and band-limited in the LCT domain; which has been shown [12] to be have an LCT defined on a discrete domain. For this special case we deduce the perfect interpolation formula for band-limited signal with discrete band limited LCT based on the sampling theorem and the LCT. To simplify the problem, in this subsection and Section IV-C and II we normalize the sampling interval as $T = \pi b / \Omega_A = 1$ and substitute it in (5) to obtain the following reconstructed signal from infinite sampling points:

$$f(t) = e^{-j(a/2b)t^2} \sum_{n=-\infty}^{\infty} e^{j(an^2/2b)} f(n) \frac{\sin(\pi t - n\pi)}{\pi t - n\pi}. \quad (22)$$

At the same time, the following equation holds for any t

$$\sin(\pi t - n\pi) = (-1)^n \sin(\pi t). \quad (23)$$

Substituting (23) into (22), results in

$$f(t) = e^{-j(a/2b)t^2} \frac{\sin(\pi t)}{\pi} \sum_{n=-\infty}^{+\infty} e^{j(an^2/2b)} f(n) \frac{(-1)^n}{t - n}. \quad (24)$$

A signal $f(t)$ said to be a *chirp-periodic* signal with period $T_A = NT$ associated with the LCT is similar to chirp-periodicity definition of a signal in the FrFT domain [42], that is to say the following equation is true for the signal samples $f(nT)$

$$e^{j(n^2/2)(a/b)} f(n) = e^{j((n+N)^2/2)(a/b)} f(n+N). \quad (25)$$

From [33], (25) can be rearranged as

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{\pi} \sum_{k=-\infty}^{+\infty} \sum_{n=-L}^{M-1} e^{j((n+kN)^2/2)(a/b)} \\ &\times f(n+kN) \frac{(-1)^{n+kN}}{t - n - kN} \end{aligned} \quad (26)$$

where L , M , and N are arbitrary integers that satisfies the relation $L + M = N$.

The inner summation in (26) can be separated for even and odd N , in detail, (26) can be rewritten as

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{\pi} \\ &\times \sum_{k=-\infty}^{+\infty} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} \\ &\times f(n) \frac{(-1)^n}{t - n - kN} \text{ for } N \text{ even} \end{aligned} \quad (27a)$$

and

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{\pi} \\ &\times \sum_{k=-\infty}^{+\infty} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} \\ &\times f(n) \frac{(-1)^n (-1)^k}{t - n - kN} \text{ for } N \text{ odd.} \end{aligned} \quad (27b)$$

Exchanging the summation in the above equation, we obtain

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{\pi} \\ &\times \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} f(n) (-1)^n \\ &\times \sum_{k=-\infty}^{+\infty} \frac{1}{t - n - kN} \text{ for } N \text{ even} \end{aligned} \quad (28a)$$

and

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{\pi} \\ &\times \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} f(n) (-1)^n \\ &\times \sum_{k=-\infty}^{+\infty} \frac{(-1)^k}{t - n - kN} \text{ for } N \text{ odd.} \end{aligned} \quad (28b)$$

Note that the two inner summations in (28) have been decomposed into partial fractions of cotangent and cosecant, respectively. Therefore, it is simple to rewrite the above equation as

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{N} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} \\ &\times f(n) (-1)^n \cot\left(\pi \frac{t-n}{N}\right) \text{ for } N \text{ even} \end{aligned} \quad (29a)$$

and

$$\begin{aligned} f(t) &= e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{N} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} \\ &\times f(n) (-1)^n \csc\left(\pi \frac{t-n}{N}\right) \text{ for } N \text{ odd.} \end{aligned} \quad (29b)$$

Equation (29) provides two interpolation formulas for recovering chirp-periodic continuous band-limited signals in the LCT

domain from even or odd number of samples. If we use some trigonometric relations, it is possible to show that the above equations can be integrated into one equation as

$$f(t) = e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{2N} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} f(n)(-1)^n \times \left[(-1)^{N+1} \tan\left(\pi \frac{t-n}{2N}\right) + \cot\left(\pi \frac{t-n}{2N}\right) \right]. \quad (30)$$

Equation (30) can be seen as the interpolation formula for band-limited and chirp-periodic signal in the time domain from finite samples associated with the LCT. We therefore present it as Theorem 6.

Theorem 6: If a chirp-periodic signal $f(t)$ with period $T_\alpha = NT$, is band limited in the LCT domain such that $L_f^A(u) = 0$ for $|u| > \Omega_A$, then the original signal $f(t)$ can be reconstructed from the finite uniformly sampled points $\tau_n = nT = n$, and the interpolation formula can be presented as

$$f(t) = e^{-j(t^2/2)(a/b)} \frac{\sin(\pi t)}{2N} \sum_{n=-L}^{M-1} e^{j(n^2/2)(a/b)} f(n)(-1)^n \times \left[(-1)^{N+1} g \tan\left(\pi \frac{t-n}{2N}\right) + \cot\left(\pi \frac{t-n}{2N}\right) \right] \quad (31)$$

where $L + M = N$.

Corollary 7: If the parameter of the LCT reduces to $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, then (31) reduces to the interpolation formula from finite uniform sampling points in the FrFT domain as follows:

$$f(t) = e^{-j(t^2/2) \cot \alpha} \frac{\sin(\pi t)}{2N} \sum_{n=-L}^{M-1} e^{j(n^2/2) \cot \alpha} f(n)(-1)^n \times \left[(-1)^{N+1} \tan\left(\pi \frac{t-n}{2N}\right) + \cot\left(\pi \frac{t-n}{2N}\right) \right] \quad (32)$$

when $\alpha = \pi/2$, (30) reduces to the classical results in the Fourier domain [26]

$$f(t) = \frac{\sin(\pi t)}{2N} \sum_{n=-L}^{M-1} f(n)(-1)^n \times \left[(-1)^{N+1} \tan\left(\pi \frac{t-n}{2N}\right) + \cot\left(\pi \frac{t-n}{2N}\right) \right]. \quad (33)$$

In this sense, the results in [25] can be seen as a special case of our results. At the same time, it must be emphasized that the representation of periodic continuous time signals based on the classical Fourier transform are only valid for band limited signals that are equidistantly sampled in accordance with the sampling theorem, whereas our results are more suitable for non-band-limited signals in the Fourier domain, i.e., it is suitable for band-limited signals in the LCT domain.

C. Chirp-Periodic Recurrent Nonuniform Sampling Signals

The results in Section IV-A give the signal reconstruction formulas from finite uniform samples associated with chirp-periodic band limited signal in the LCT domain, whereas in the following subsections the signal reconstruction formula from finite recurrent nonuniform sampling points is deduced as in Theorem 8.

Theorem 8: Let $f(t)$ be a Ω_A band limited signal in the LCT domain, suppose it is also chirp-periodic with period $T_A = NT = b\pi N/\Omega_A$, then $f(t)$ can be perfectly reconstructed from its finite recurrent nonuniformly samples $f(t_p + mNb\pi/2\Omega_A)$, where $p = 1, \dots, N$, $m = \dots, -1, 0, 1, \dots$, and the reconstruction formula is

$$f(t) = e^{-j(a/2b)t^2} \sum_{p=1}^N f(t_p) e^{j(a/2b)t_p^2} \psi_p(t) \quad (34)$$

where

$$\psi_p(t) = \begin{cases} \frac{\cos\left(\pi \frac{t-t_p}{N}\right) \prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t-t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t_p-t_q)}, & N \text{ even} \\ \frac{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t-t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t_p-t_q)}, & N \text{ odd.} \end{cases} \quad (35)$$

Proof: From Theorem 3 the signal can be reconstructed via the following formula from the recurrent nonuniform samples $\tau_{pm} = t_p + mb\pi N/\Omega_A = t_p + mNT$, $p = 0, 1, \dots, N-1$, $m = \dots, -1, 0, 1, \dots$:

$$f(t) = e^{-j(a/2b)t^2} \sum_{m=-\infty}^{\infty} \sum_{p=1}^N f(\tau_{pm}) e^{j(a/2b)\tau_{pm}^2} \times \frac{\prod_{q=1}^N \sin \frac{\pi \Omega_A}{bN}(t-t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi \Omega_A}{bN}(t_p-t_q)} \frac{(-1)^{mN}}{\frac{\pi \Omega_A}{bN}(t-t_p - mN \frac{\pi b}{\Omega_A})}. \quad (36)$$

If we normalize the sampling interval as $T = \pi b/\Omega_A = 1$, then (36) reduces to

$$f(t) = e^{-j(a/2b)t^2} \sum_{m=-\infty}^{\infty} \sum_{p=1}^N f(\tau_{pm}) e^{j(a/2b)\tau_{pm}^2} \times \frac{\prod_{q=1}^N \sin \frac{\pi}{N}(t-t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t_p-t_q)} \frac{(-1)^{mN}}{\frac{\pi}{N}(t-t_p - mN)}. \quad (37)$$

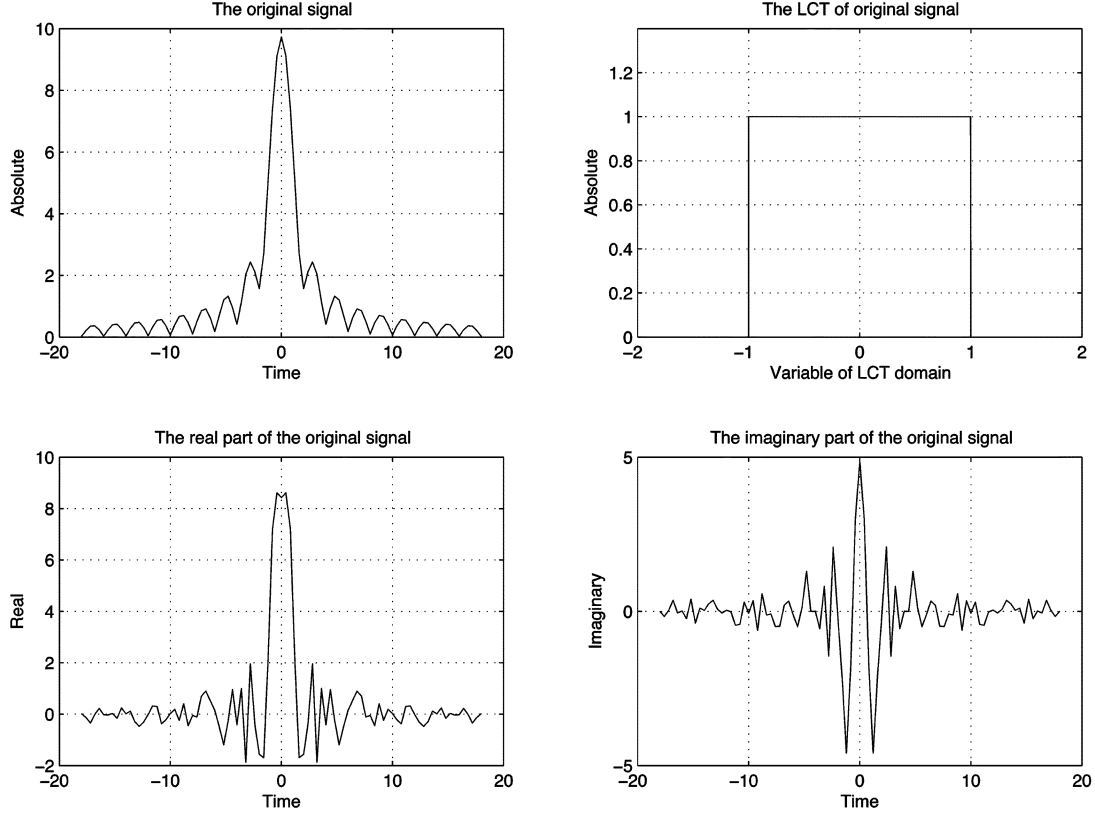


Fig. 4. Original signal in the time domain and the LCT domain.

Exchanging the summation of (37), we obtain

$$f(t) = e^{-j(a/2b)t^2} \sum_{p=1}^N f(\tau_{pm}) e^{j(a/2b)\tau_{pm}^2} \times \frac{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t - t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t_p - t_q)} \sum_{m=-\infty}^{\infty} \frac{(-1)^{mN}}{\frac{\pi}{N}(t - t_p - mN)}. \quad (38)$$

Since $f(t)$ is chirp-periodic with period $T_\alpha = NT = b\pi N/\Omega_A$, (38) reduces to

$$f(t) = e^{-j(a/2b)t^2} \sum_{p=1}^N f(t_p) e^{j(a/2b)t_p^2} \times \frac{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t - t_q)}{\prod_{q=1, q \neq p}^N \sin \frac{\pi}{N}(t_p - t_q)} \sum_{m=-\infty}^{\infty} \frac{(-1)^{mN}}{\frac{\pi}{N}(t - t_p - mN)}. \quad (39)$$

One can prove the final result by applying the methods in Theorem 5 to (38). ■

V. SIMULATION RESULTS AND POTENTIAL APPLICATION

In order to show the importance of the LCT sampling theory and the correctness of the results, we choose the recurrent nonuniform sampling model as an example to perform the simulation in the LCT domain; the potential application of the results in the A/D converter performance analysis is also proposed to show the advantage of the theory.

A. The Simulation Results for the Recurrent Nonuniform Sampling

In this subsection, we choose the recurrent nonuniform sampling as an example to verify the correctness of the results. A real rectangular function in the LCT domain can be seen as a band-limited signal. Fig. 4 shows the LCT pair of this signal with parameter $\mathbf{A} = \begin{pmatrix} 1+2/\pi & 2/\pi \\ 1 & 1 \end{pmatrix}$, where the LCTed signal is a real rectangular function confined to the interval $(-1, 1)$.

According to Lemma 2, the signal in Fig. 4 can not be band-limited in the Fourier domain, in other words, applying the Fourier domain based methods [26]–[30] to this signal, may lead to wrong (or at least suboptimal) reconstruction results. In order to reconstruct the original signal perfectly, the LCT is applied and the process of the reconstruction from the periodic nonuniform sampling points can be described as following.

In fact, from the uniform sampling theory of the LCT domain [14], [15], the uniform sampling period can be chosen as $T = 2$; if we let the recurrent period $N = 3T = 6$ and the sampling point in the first recurrent period be $[t_1 = 0, t_1 = 2.2, t_1 =$

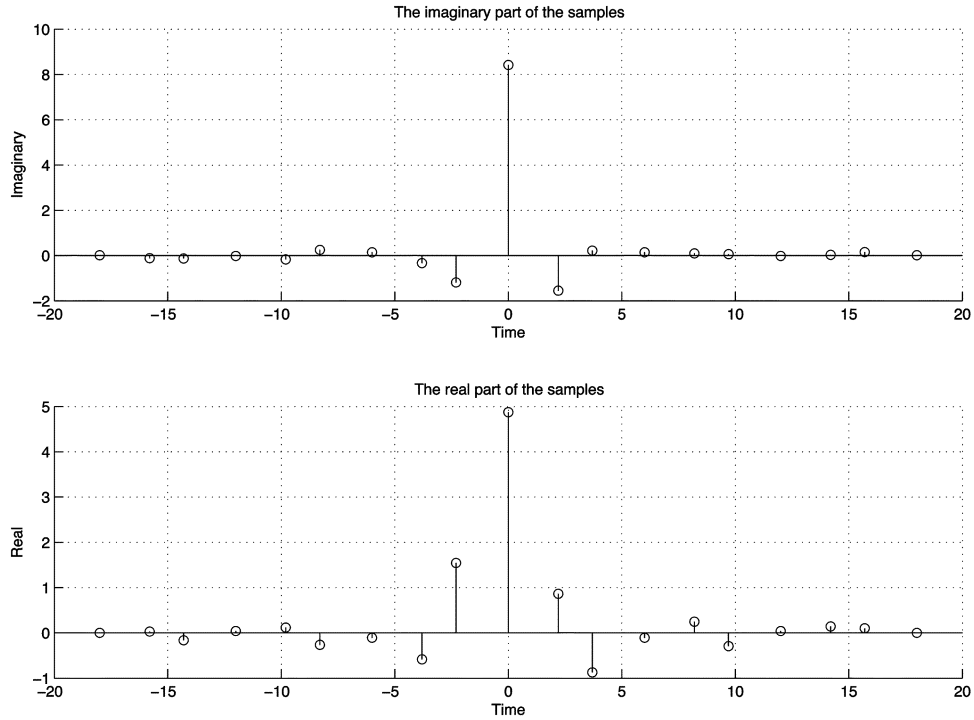


Fig. 5. Recurrent nonuniform sampling set.

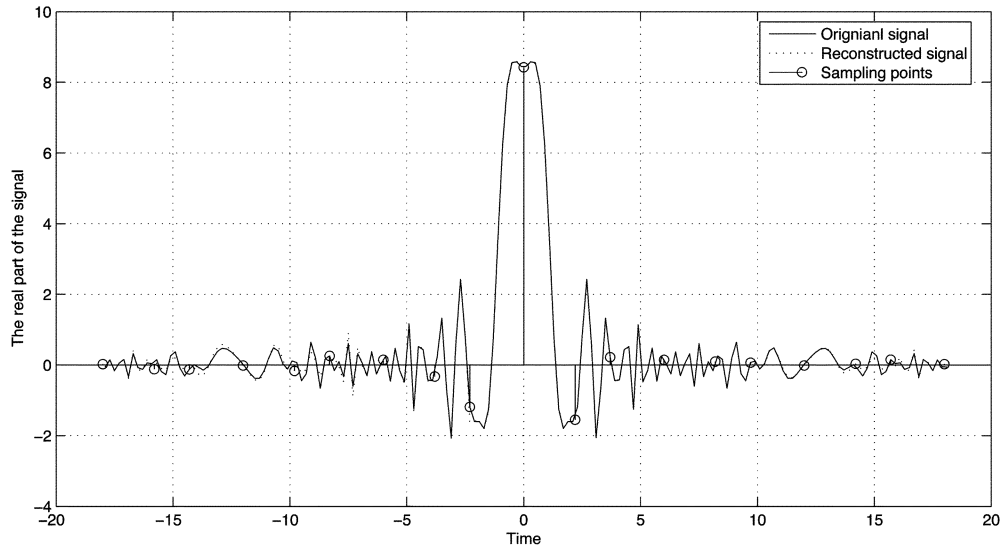


Fig. 6. Real parts of the original, sampling points and the reconstructed signals.

3.7], then the whole sampling set can be described as $\tau_{pm} = t_p + m \times 3T$, $p = 1, 2, 3$, $m = \dots, -1, 0, 1, \dots$. The real and imaginary part of the sampling set $(\tau_{pm}, f(\tau_{pm}))$ is also plotted in Fig. 5.

From the reconstruction formula of the recurrent nonuniform sampling model derived in Theorem 1, we can reconstruct the original signal. Figs. 6 and 7 plot the real and imaginary parts of the original, sampling points and the reconstructed signals, respectively.

B. The Potential Application

The sampling process is one of the fundamental topics in the signal processing community, therefore deriving the reconstruc-

tion formulas from uniform or nonuniform sampling points associated with the LCT can be seen as the first step in the application of the LCT in real world. In this subsection, we will briefly discuss the potential applications of the results derived in the manuscript in the LCT domain.

The derived results and the simulation reveal that the immediate application of the results can be found in the reconstruction of the original signal from the nonuniform sampling points. For example, a practical application for periodic nonuniform sampling points can be found in the area of A/D converters. Other potential applications can be found in the sequential sampling optical fields with subpixel translated optical arrays to obtain “superresolution,” where every first order optical system can be

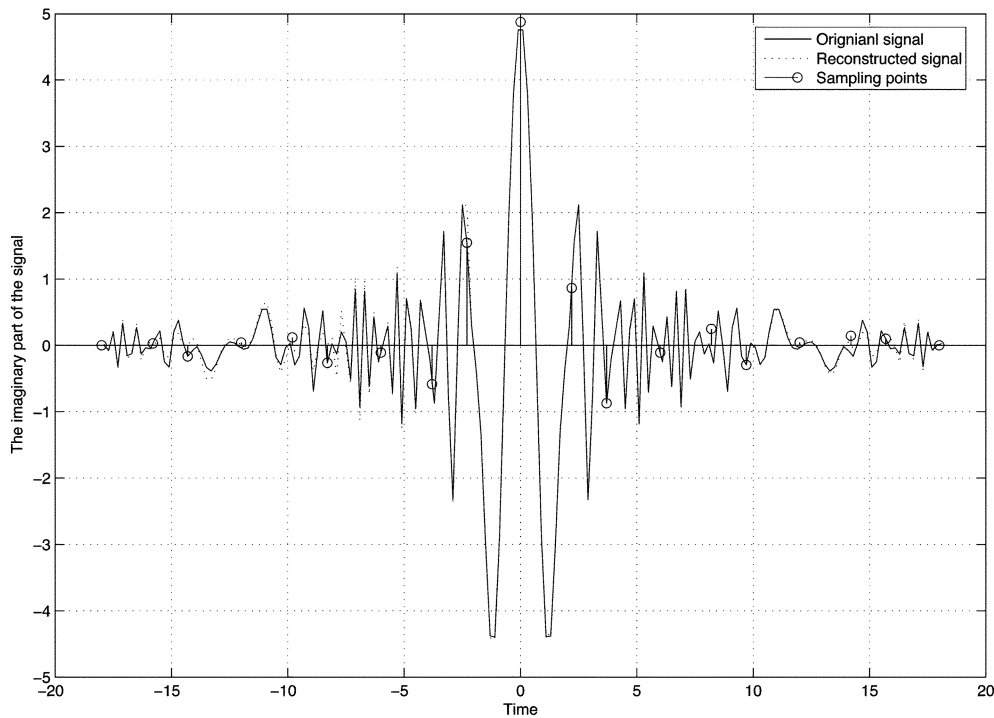


Fig. 7. Imaginary parts of the original, sampling points and the reconstructed signals.

modeled by an LCT; fast sampling of dispersed pulse, where dispersion in an optical communication channel can be described using LCT; sampling of the radar chirped echo and acoustical echoes, where the better results can be derived by LCT. In these cases, remarkable results can be achieved by the use of the LCT instead of the FT for the reconstruction of the signal using the uniform or nonuniform signal samples. How to derive better results using the LCT in real applications are also our future research directions.

VI. CONCLUSION

Focusing on the existing sampling problems associated with the LCT, this paper investigates the nonuniform sampling and the signal reconstruction formulas in the LCT domain. We first examine some special nonuniform sampling processes in detail, deduce the reconstruction formulas from these nonuniformly sampling points associated with the LCT and then resolve the practical implementation problem of the sampling formulas for chirp-periodic band-limited signals from finite uniform or nonuniform samples in the LCT domain. The simulation results and the potential applications of the results are also presented. In addition, the results of the paper can be seen as the extension of the classical results in the Fourier domain to the LCT domain.

REFERENCES

- [1] I. Daubechies, *Ten Lectures on Wavelets*. Philadelphia, PA: SIAM, 1992.
- [2] V. Namias, "The fractional order Fourier transform and its application to quantum mechanics," *J. Inst. Math. Appl.*, vol. 25, pp. 241–265, 1980.
- [3] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, *The Fractional Fourier Transform with Applications in Optics and Signal Processing*. New York: Wiley, 2001.
- [4] R. Tao, L. Qi, and Y. Wang, *Theory and Applications of the Fractional Fourier Transform*. Beijing, China: Tsinghua Univ. Press, 2004.
- [5] M. Moshinsky and C. Quesne, "Linear canonical transformations and their unitary representations," *J. Math. Phys.*, vol. 12, no. 8, pp. 1772–1783, Aug. 1971.
- [6] L. M. Bernardo, "ABCD matrix formalism of fractional Fourier optics," *Opt. Eng.*, vol. 35, no. 3, pp. 732–740, Mar. 1996.
- [7] S. C. Pei and J. J. Ding, "Eigenfunctions of linear canonical transform," *IEEE Trans. Signal Process.*, vol. 50, no. 1, pp. 11–26, Jan. 2002.
- [8] D. F. V. James and G. S. Agarwal, "The generalized Fresnel transform and its applications to optics," *Opt. Commun.*, vol. 126, no. 5, pp. 207–212, May 1996.
- [9] M. J. Bastiaans, "Wigner distribution function and its application to first order optics," *J. Opt. Soc. Amer.*, vol. 69, pp. 1710–1716, 1979.
- [10] M. Nazarathy and J. Shamir, "First-order optics-A canonical operator representation: Lossless systems," *J. Opt. Soc. Amer.*, vol. 72, pp. 356–364, 1982.
- [11] L. Onural, A. Gotchev, H. M. Ozaktas, and E. Stoykova, "A survey of signal processing problems and tools in holographic three-dimensional television," *IEEE Trans. Circuit Syst. Video Technol.*, vol. 17, no. 11, Nov. 2007.
- [12] A. Stern, "Why is the linear canonical transform so little known?," *Proc. AIP*, vol. 860, pp. 225–234, Jun. 2006.
- [13] M. J. Hackert, "Explanation of launch condition choice for GRIN multimode fiber attenuation and bandwidth measurements," *J. Lightw. Technol.*, vol. 10, pp. 125–129, Feb. 1992.
- [14] A. Stern, "Sampling of linear canonical transformed signals," *Signal Process.*, vol. 86, no. 7, pp. 1421–1425, Jul. 2006.
- [15] B.-Z. Li, R. Tao, and Y. Wang, "New sampling formulae related to linear canonical transform," *Signal Process.*, vol. 87, no. 5, pp. 983–990, May 2007.
- [16] A. Stern, "Sampling of compact signals in offset linear canonical transform domains," *Signal, Image, Video Process.*, vol. 1, no. 4, pp. 359–367, Oct. 2007.
- [17] E. Van Der Ouderaa and J. Renneboog, "Some formulas and applications of nonuniform sampling of bandwidth-limited signals," *IEEE Trans. Instrum. Meas.*, vol. 37, pp. 353–357, Sep. 1988.
- [18] P. J. S. G. Ferreira, "Sampling and generalized almost periodic extension of functions," *IEEE Trans. Circuits Syst. II*, vol. 46, no. 4, pp. 475–578, Apr. 1999.
- [19] F. Gori, "Sampling in optics," in *Advanced Topics in Shannon Sampling and Interpolation Theory*, R. J. Marks, II, Ed. New York: Springer-Verlag, 1993, pp. 37–83.

- [20] H. Stark, "Polar, spiral, and generalized sampling and interpolation," in *Advanced Topics in Shannon Sampling and Interpolation Theory*, R. J. Marks, II, Ed. New York: Springer-Verlag, 1993, pp. 185–218.
- [21] G. J. Marseille, M. Fuderer, R. de Beer, A. F. Mehlkopf, and D. van Ormondt, "Reduction of MRI scan time through nonuniform sampling and edge-distribution modeling," *J. Magn. Reson. B*, vol. 103, pp. 292–295, 1994.
- [22] R. G. Wiley, H. Schwarzlander, and D. D. Weiner, "Demodulation procedure for very wideband FM," *IEEE Trans. Commun.*, vol. COM-25, pp. 318–327, Mar. 1977.
- [23] M. Piacentini, C. Cafforio, and F. Rocca, "Algorithms for image reconstruction after nonuniform sampling," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-35, pp. 1185–1189, Aug. 1987.
- [24] R. R. Scott, Jr and G. S. Smith, "Error corrections for an automated time-domain network analyzer," *IEEE Trans. Instrum. Meas.*, vol. IM-35, pp. 300–303, Sep. 1986.
- [25] P. K. Murphy and N. C. Gallagher, "Hexagonal sampling techniques applied to Fourier and Fresnel digital holograms," *J. Opt. Soc. Amer.*, vol. 72, no. 7, pp. 929–937, Jul. 1982.
- [26] A. J. Jerri, "The Shannon sampling theorem-its various extensions and applications: A tutorial review," *Proc. IEEE*, vol. 65, pp. 1565–1596, Nov. 1977.
- [27] Y. C. Eldar and A. V. Oppenheim, "Filterbank reconstruction of bandlimited signals from nonuniform and generalized samples," *IEEE Trans. Signal Process.*, vol. 48, no. 10, pp. 2864–2875, Oct. 2000.
- [28] A. Feuer and G. C. Goodwin, "Reconstruction of multidimensional bandlimited signals from nonuniform and generalized samples," *IEEE Trans. Signal Process.*, vol. 53, pp. 4273–4282, Nov. 2005.
- [29] Y. C. Jenq, "Digital spectra of nonuniformly sampled signals: Fundamentals and high-speed waveform digitizers," *IEEE Trans. Instrum. Meas.*, vol. 37, pp. 245–251, Jun. 1988.
- [30] Y. C. Jenq and L. Cheng, "Digital spectrum of a nonuniformly sampled two-dimensional signal and its reconstruction," *IEEE Trans. Instrum. Meas.*, vol. 54, pp. 1180–1187, Jun. 2005.
- [31] R. Tao, B.-Z. Li, and Y. Wang, "Spectral analysis and reconstruction for periodic nonuniformly sampled signals in fractional Fourier domain," *IEEE Trans. Signal Process.*, vol. 55, pp. 3541–3547, Jul. 2007.
- [32] M. R. Belmont, "Nonuniform sampling specifically for finite-length data," *Proc. Inst. Elect. Eng.—F*, vol. 140, no. 1, pp. 55–62, Feb. 1993.
- [33] T. Schanze, "Sinc interpolation of discrete periodic signals," *IEEE Trans. Signal Process.*, vol. 43, no. 6, pp. 1502–1503, Jun. 1995.
- [34] X. Xia, "On bandlimited signals with fractional Fourier transform," *IEEE Signal Process. Lett.*, vol. 3, pp. 72–74, Mar. 1996.
- [35] J. J. Haelly and J. T. Sheridan, "Cases where the linear canonical transform of a signal has compact support or is band-limited," *Opt. Lett.*, vol. 33, no. 9, pp. 228–230, Jan. 2008.
- [36] K. Yao and J. B. Thomas, "On some stability and interpolatory properties of nonuniform sampling expansions," *IEEE Trans. Circuit Theory*, vol. CT-14, pp. 404–408, Dec. 1967.
- [37] A. Papoulis, "Generalized sampling expansion," *IEEE Trans. Circuit Syst.*, vol. CAS-24, no. 11, pp. 652–654, Nov. 1977.
- [38] F. Marvasti, "Nonuniform sampling," in *Advanced Topics in Shannon Sampling and Interpolation Theory*, R. J. Marks, II, Ed. New York: Springer-Verlag, 1993, pp. 121–156.
- [39] J. L. Yen, "On nonuniform sampling of bandwidth-limited signals," *IEEE Trans. Circuit Theory*, vol. 3, pp. 251–257, Dec. 1956.
- [40] A. I. Zayed, *Advances in Shannon's Sampling Theory*. Boca Raton, FL: CRC Press, 1993.
- [41] H. D. Helms and J. B. Thomas, "Truncation error of sampling theorem expansion," *Proc. IRE*, vol. 50, pp. 179–184, 1962.
- [42] T. Erseghe, P. Kraniakaskas, and G. Cariolaro, "Unified fractional Fourier transform and sampling theorem," *IEEE Trans. Signal Process.*, vol. 47, no. 12, pp. 3419–3423, Dec. 1999.



Ran Tao (M'00–SM'04) received the Ph.D. degree in electrical engineering from Harbin Institute of Technology, Harbin, in 1993.

He has been a Professor at the Beijing Institute of Technology, China, since 1999. From March 2001 to April 2002, he was a visiting scholar in the University of Michigan, Ann Arbor. His research interests are in the areas of the fractional Fourier transform, time-frequency signal processing, and signal processing for radar and communications.

Dr. Tao was the recipient of National Science Foundation of China for Distinguished Young Scholars in 2006, and of Teaching and Research Award for Outstanding Young Teachers in Higher Education Institutions of the Ministry of Education, China, in 2000. He is the Vice-President of Chinese Radar Industry Association and a Fellow of Chinese Institute of Electronics.



Bing-Zhao Li was born in Yi Nan County, Shandong Province, China, in 1975. He received the B.S. degree in mathematics from Shandong Normal University, Jinan, China, in 1998 and the M.S. and Ph.D. degrees from the Beijing Institute of Technology (BIT), Beijing, China, in 2001 and 2007, respectively.

Presently, he is an Associate Professor with the Mathematics Department, BIT. His research interests include time-frequency analysis, digital signal processing, and image processing.



Yue Wang received the B.S. degree in radar engineering from Xidian University, Shanxi, China, in 1956.

He was the President of Beijing Institute of Technology (BIT), China, from 1993 to 1997. He is currently a Professor with BIT. His research interests include information system theory and technology, and signal processing.

Prof. Wang is a Fellow of the Chinese Academy of Sciences and also a Fellow of the Chinese Academy of Engineering.



George Kwamina Aggrey received the M.Sc. degree in telecommunication engineering from the Moscow Technical University of Communication and Informatics. He is currently working towards the Ph.D. degree in the Department of Electrical and Electronics Engineering under the supervision of Prof. Ran Tao at the Beijing Institute of Technology, Beijing, China.

From 2004 to 2006, he joined Ghana Telecom Company Limited. During his stay with Ghana Telecom, he was also a part-time Lecturer. His current research areas include sampling, time-frequency representations, fractional Fourier transform, and transmultiplexers.